VEPOLINK
MONITORING PURITY

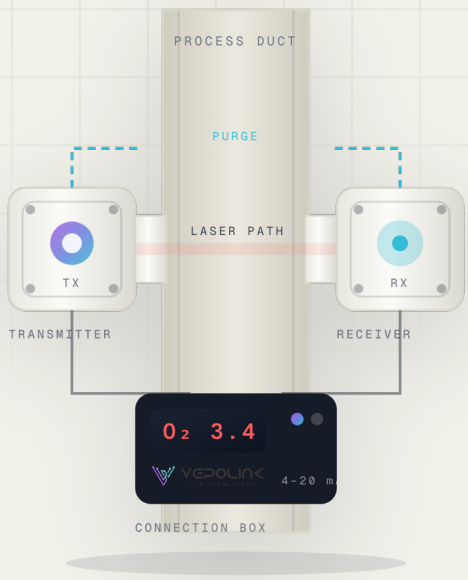
ENVIRONMENTAL INTELLIGENCE × PROCESS ANALYTICS

CROSS DUCT GAS ANALYZER · TDLAS · IN-SITU

Cross Duct Gas Analyzer

Measure it where it flows

A flameproof tunable diode laser analyzer that reads O_2, CO, NH_3, CO_2, CH_4, H_2O, HCl and HF straight across the duct — no sampling line, no conditioning, no lag between the process and the number.



20 m
OPTICAL CHANNEL LENGTH

1 s
RESPONSE TIME

8 gases
O_2 · CO · NH_3 · CO_2 · CH_4 · H_2O · HCL · HF

EX D IIC T6 GB FLAMEPROOF | **IP66** DUST & WATER | **TDLAS** SINGLE-LINE

1 s	1 % FS	1 % FS	2 ×/yr
RESPONSE TIME	LINEARITY ERROR	ZERO & SPAN DRIFT PER HALF YEAR	MAINTENANCE & CALIBRATION

01 OVERVIEW

The sample never *leaves the duct.*

Extractive analyzers spend most of their life fighting the sample: heated lines that block, filters that load, coolers that strip the very component you wanted to measure. A cross duct analyzer removes all of it. The laser crosses the duct, the gas absorbs, and the concentration is known — in situ, in about a second.

The Vepolink Cross Duct Gas Analyzer is a flameproof tunable laser analyzer built on TDLAS. It comprises a transmitter, a receiver and a connection box: the transmitter drives a tunable diode laser at a wavelength specific to the target gas, the receiver performs photoelectric conversion, signal processing and spectral analysis, and the connection box brings power and signals out to the plant. Because the measurement is optical and the spectral line is unique to the gas, background components in the flue do not interfere.

AT A GLANCE

- TDLAS
- IN-SITU
- SINGLE-LINE SPECTRUM
- UP TO 20 M PATH
- 4-20 MA
- RS-485 · RS-232 · GPRS
- EX D IIC T6 GB
- IP66

· PRODUCT FEATURES

Easy Optical Alignment

XY-direction adjustment on the mounting brings the beam onto the receiver quickly at commissioning and holds it through thermal movement of the duct.

Single-Line Spectrum

The analyzer locks onto one absorption line of the target gas, so water, CO_2 and other background species in the flue do not bias the reading.

Integrated Flameproof Build

Ex d IIC T6 Gb construction with IP66 sealing in one integrated structure — high stability, high reliability, suitable for hazardous process areas.

Low Drift, Long Intervals

1 % F.S. zero and span drift per half year, with two clean-and-calibrate visits a year. There is no pretreatment system to service in between.

02 MEASUREMENT PRINCIPLE

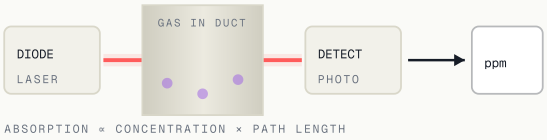
Light in, *concentration out.*

Tunable Diode Laser Absorption Spectroscopy measures how much light a gas removes from a beam at one precise wavelength. Everything else in the instrument exists to make that measurement stable in a hot, dusty, moving duct.

TDLAS

Absorption Across the Duct

DIODE LASER · OPTICAL PATH UP TO 20 M



The transmitter drives a tunable diode laser to emit a specific wavelength through the measurement environment. Target molecules absorb part of that light; the attenuation at the line centre gives concentration directly.

`Absorption ∝ concentration × path`

SELECTIVITY

One Line, No Interference

O_2 · CO · NH_3 · CO_2 · CH_4 · H_2O · HCL · HF



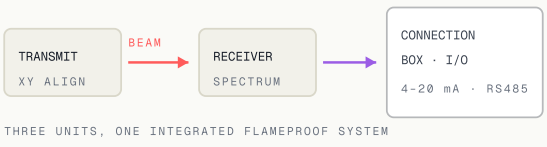
Scanning a single, isolated absorption line rather than a broad band minimises interference from background gases — the reason a laser analyzer can work in raw flue gas without conditioning.

`Single-line spectrum technology`

SYSTEM

Transmitter · Receiver · Box

PHOTOELECTRIC CONVERSION · SPECTRAL ANALYSIS



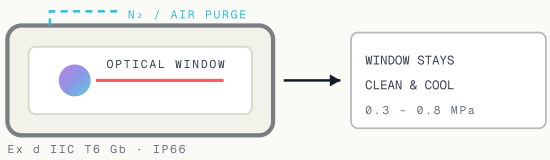
The receiver converts the transmitted light to a signal, processes it and analyses the spectrum to determine concentration. The connection box provides convenient interfaces for power and signal wiring.

`Three units, one instrument`

PROTECTION

Flameproof & Purged

EX D IIC T6 GB · IP66 · 0.3-0.8 MPA PURGE



An integrated flameproof structure allows installation in hazardous areas, and a nitrogen or instrument-air purge keeps the optical windows clean — using less gas than positive-pressure designs.

`Lower purge consumption`

03 MEASURED GASES

Eight gases, *one optical path.*

The analyzer performs in-situ analysis of the gases below under various complex industrial conditions. Measuring range and resolution are configured per application — tell us the duct, the temperature and the expected concentration and we confirm the configuration at order.

GAS	NAME	TYPICAL SERVICE
O_2	Oxygen	Combustion control, excess air trim
CO	Carbon Monoxide	Incomplete combustion, safety
NH_3	Ammonia	SCR / SNCR ammonia slip
CO_2	Carbon Dioxide	Process and emission monitoring

GAS	NAME	TYPICAL SERVICE
CH_4	Methane	Process gas, leak and loss control
H_2O	Moisture	Wet-basis correction, drying control
HCl	Hydrogen Chloride	Waste incineration, scrubber control
HF	Hydrogen Fluoride	Aluminium, glass, brick kilns

STEP 01

Flange

Opposed flanges are fitted on the duct wall and the transmitter and receiver mounted in situ across the process stream.

STEP 02

Align

XY adjustment brings the beam onto the receiver aperture; signal strength is checked at the connection box.

STEP 03

Purge

Industrial nitrogen or purified instrument air at 0.3–0.8 MPa keeps the optical windows clear of dust and condensate.

STEP 04

Compensate

Two 4–20 mA inputs accept duct temperature and pressure so concentration is reported on a corrected basis.

Because the measurement is in-situ, there is no sample pretreatment system: nothing to adsorb the sample, nothing to block, and no risk of altering the gas between the duct and the detector. Response is about one second, so the reading tracks the process rather than reporting on it a minute later.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

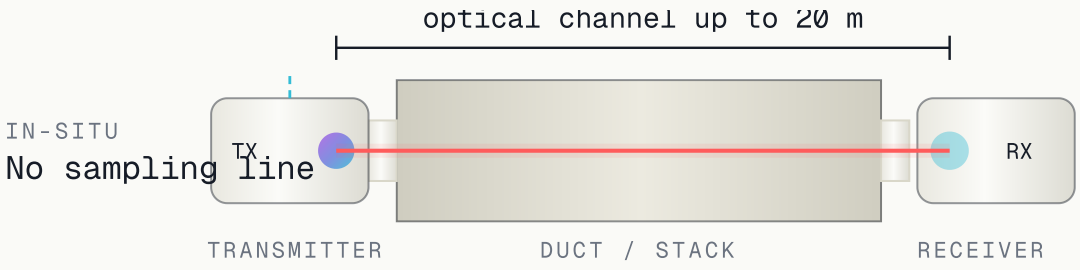
Measured Gases	O_2, CO, NH_3, CO_2, CH_4, H_2O, HCl, HF
Technology	TDLAS tunable diode laser
Measurement Type	In-situ, cross duct
Optical Channel Length	20 m
Linearity Error	1 % F.S.
Repeatability	1 %
Zero Drift	1 % F.S. per half year
Span Drift	1 % F.S. per half year
Response Time	1 s
Maintenance Cycle	2 times/year, clean optical window
Calibration Cycle	2 times/year

OUTPUT · UTILITIES · RATING

Analog Output	2 × 4-20 mA
Analog Input	2 × 4-20 mA, temperature & pressure compensation
Digital Output	RS-485 · RS-232 · GPRS
Relay Output	2 relays, 24 V, 1 A
Power Supply	24 V DC, < 20 W
Purge Gas	0.3-0.8 MPa industrial nitrogen or purified instrument air
Type of Protection	Ex d IIC T6 Gb
Enclosure Rating	IP66
Ambient Temperature	-20 °C to +60 °C
Installation	In-situ, opposed flanges on duct
System	Transmitter · receiver · connection box

INSTALLATION GEOMETRY

OPPOSED FLANGE MOUNTING · OPTICAL CHANNEL UP TO 20 M



05 INSTALLATION & UPKEEP

Aligned once, *left to work.*

The measurement section decides the result. Choose a straight run with **representative, well-mixed flow**, away from bends, dampers and air ingress; keep the two flanges **truly opposed** so alignment survives thermal movement; and give both heads a **steady purge** so the optical windows stay clean between the two scheduled visits a year.

01 · LOCATION

Representative section

A straight length of duct with well-mixed flow and stable temperature, clear of bends, dampers and leakage points.

02 · FLANGES

Opposed pair

Matching flanges welded on opposite duct walls, concentric on the optical axis, carrying the transmitter and the receiver.

03 · ALIGNMENT

XY adjustment

The mounting adjusts in X and Y to centre the beam on the receiver aperture; signal level confirms a good optical path.

04 · PURGE

N_2 or instrument air

0.3–0.8 MPa industrial nitrogen or purified instrument air protects the windows and consumes less than positive-pressure designs.

05 · WIRING

Connection box

24 V DC in at under 20 W; two 4–20 mA outputs, two compensation inputs, RS-485/RS-232/GPRS and two relays out.

06 · SERVICE

Twice a year

Clean the optical windows and calibrate — two visits a year. There is no sample line, filter or cooler to maintain.

IN-SITU VERSUS EXTRACTIVE

An extractive system must pull gas out, keep it hot, dry it, filter it and deliver it unchanged — a chain in which adsorption, blockage and sample damage are routine failure modes. The cross duct analyzer measures in place, so that chain does not exist: less to install, less to go wrong, and a reading that reflects the duct as it is now.

COMPENSATION AND REPORTING

Duct temperature and pressure transmitters feed the two 4–20 mA inputs so concentration is corrected to the reporting basis your authority requires. Outputs go to DCS or PLC as 4–20 mA, to a data system over RS-485 or RS-232, and to a remote server over GPRS where no cable run exists.

06 APPLICATIONS

Where it *works.*

Wherever the gas is too hot, too wet or too dirty to extract — and wherever the answer is needed in seconds rather than minutes.

Industrial Emission Monitoring	Furnace Gas Monitoring	Boiler Gas Monitoring
Stack Gas Monitoring	Duct Gas Monitoring	Continuous Emission Monitoring
Refineries & Petrochemicals	Cement, Steel & Kilns	Waste Incineration & SCR

PROCESS CONTROL, NOT JUST REPORTING

A one-second in-situ reading is fast enough to close a loop. Oxygen across a boiler duct trims excess air; ammonia after an SCR governs reagent dosing; CO warns of incomplete combustion before it becomes a loss or a hazard. The same instrument that satisfies the emission record also pays for itself in fuel and reagent.

SPECIFYING A MEASUREMENT POINT

We need the gas, the duct dimension, the process temperature and pressure, the dust loading and the area classification. From that we confirm the optical path, the purge requirement, the flange design and the output configuration as one package.

07 FAQ

Frequently asked *questions.*

What is the Cross Duct Gas Analyzer?

An industrial gas analyzer using TDLAS technology for direct in-situ measurement of gases in ducts and large process streams, mounted across the duct on opposed flanges.

Which gases can it measure?

O_2, CO, NH_3, CO_2, CH_4, H_2O, HCl and HF. The configuration is set for the target gas and the process conditions at order.

What technology does it use?

Tunable Diode Laser Absorption Spectroscopy — a tunable diode laser scanned across a single absorption line of the target gas.

Does it need a sample pretreatment system?

No. Measurement is in-situ, which avoids a conventional pretreatment system and the adsorption, blockage and sample damage that come with it.

What is the response time?

Approximately one second, fast enough to use the reading for combustion or reagent control rather than reporting alone.

How long an optical path is supported?

Up to a 20 m optical channel length, covering most duct and stack diameters in power, cement, steel and chemical plants.

Is it suitable for hazardous areas?

Yes. The integrated flameproof structure is certified Ex d IIC T6 Gb with an IP66 enclosure rating.

What purge gas is required?

Industrial nitrogen or purified instrument air at 0.3–0.8 MPa. Consumption is lower than positive-pressure models.

What outputs are available?

Two 4–20 mA analog outputs, two 4–20 mA inputs for temperature and pressure compensation, RS-485, RS-232, GPRS and two 24 V 1 A relays.

How much maintenance does it need?

Twice a year: clean the optical windows and calibrate. Zero and span drift are 1 % F.S. per half year.

This brochure is general product information. Configuration, measuring ranges, installation geometry and performance figures are indicative and confirmed at order against the process conditions of the measurement point. Follow the user manual, site procedures and applicable standards for installation, purging, calibration and use in hazardous areas. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

IN-SITU, IN ONE SECOND

Measure the duct, *not the sample line.*

Send us the gas, the duct size and the process conditions, and we'll confirm the optical path, purge, flanges and outputs as one package. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specs, installs and aligns cross duct laser analyzers. No call centre.
+91 99538 86178 →

02 **Request a measurement-point proposal**
RECOMMENDED
Share duct dimensions, gas, temperature, pressure and dust loading — we'll return path length, purge and output configuration.
sales@vepolink.com →

03 **Book a demonstration**
See alignment, live spectrum and output on a working analyzer with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM